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Solution-Operator Semi-Discretization: General-Order Multiplication-Free Stability Analysis of Time-Periodic Delayed Systems and the Accuracy–Sparsity Trade-Off

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Abstract

The multiplication-free semi-discretization method (MFSD) reduces the stability analysis of time-periodic delay differential equations to a sparse, banded generalized eigenvalue problem with linear time complexity in the resolution p; however, the classical piecewise-constant step limits its convergence order to two. This paper introduces the *solution-operator semi-discretization* (SOSD) method: since the multiplication-free representation of the one-period solution operator only requires a per-step residual, the integrator inside each step becomes a free design choice. The residual block rows of any explicit Runge–Kutta or implicit collocation-type scheme (Gauss–Legendre, Radau, Lobatto) are embedded in the banded system through a Kronecker-product structure of the Butcher tableau and the system matrices, raising the convergence order of the dominant characteristic multipliers far beyond the classical limit of two: the full superconvergent order $2s$ of s-stage Gauss collocation is observed when the delay is commensurate with the mesh, and orders between the interpolation-limited $s+1$ and $2s$ for general (including time-periodic) delays, in agreement with classical DDE interpolation theory. In the limit of a single ultra-high-order step, the framework degenerates to the known global spectral method; between the two extremes lies a continuum in which per-step order is traded against matrix sparsity. Because the cost at fixed step count grows polynomially with the stage number (measured exponents 1.6–2.0 for the rank-one delay couplings typical of machining and feedback systems, up to cubic for dense coupling) while the h-refinement cost is linear up to the near-constant Krylov iteration count (measured exponents 1.0–1.1), the CPU-time optimum for a prescribed accuracy is a problem-dependent *sweet spot* at moderate order and many steps: its location is measured for systems requiring only tens of points per period and for systems — with natural frequency high relative to the period — requiring many hundreds, where the banded moderate-order route wins decisively. We further argue that previous method comparisons conflated order refinement with h-refinement, and we prescribe a fair work–precision methodology (log–log axes, wide resolution ranges, repetition-robust CPU timing, a single machine and BLAS-thread configuration). All methods are implemented in the open-source Julia package SOSD.jl.

Keywords

time-delay systems, semi-discretization, solution operator, Runge–Kutta methods, Floquet theory, stability analysis, computational complexity

Introduction

Time-periodic linear delay differential equations (DDEs) of the form

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\,\mathbf{x}(t) + \sum_{k=1}^{g} \mathbf{B}_k(t)\,\mathbf{x}\big(t-\tau_k(t)\big) + \mathbf{c}(t), \quad (1)$$

with T-periodic coefficients $\mathbf{A}(t)$, $\mathbf{B}_k(t)$, delays $\tau_k(t)$ and forcing $\mathbf{c}(t)$, govern a wide range of engineering and biological systems: regenerative machine-tool chatter (Tobias 1965; Altintas 2012; Munoa et al. 2016), turning with spindle-speed variation (Long and Balachandran 2007; Seguy et al. 2008), feedback control with latency (Stépán 1989), and seasonally forced population dynamics. Their asymptotic stability is determined by the dominant characteristic multipliers of the monodromy operator associated with one period (Insperger and Stépán 2011).

Two families of numerical methods dominate practice. Semi-discretization (SD) (Insperger and Stépán 2002, 2011) advances the state with a fixed step $\Delta t = T/p$, treating the delayed term (and, in the classical zeroth- and first-order variants, the coefficients) as piecewise constant or piecewise linear per step; its convergence order is limited to two (Insperger et al. 2008). Collocation and pseudospectral methods (Breda et al. 2005, 2015) represent

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the solution segment by a global high-order polynomial and converge spectrally for smooth problems. A recurring claim in the literature is that collocation "outperforms" semi-discretization; we argue in Section that such comparisons are structurally unfair, because they compare an order-refinement strategy (p-type refinement in FEM terminology, raising the polynomial order) with an h-refinement strategy (shrinking the step) — the same category error as declaring p-type finite elements universally superior to h-type ones.

The recent multiplication-free semi-discretization method (MFSD) (Bachrathy 2026) removed the main cost bottleneck of SD: instead of multiplying p one-step matrices into a dense monodromy matrix ($\mathcal{O}(p^{2.8})$ in practice), every step is written as a residual equation and all p steps are stacked into one sparse banded system, split into the one-period mapping $\mathbf{\Phi}_{\mathrm{L}}\mathbf{y}_p = \mathbf{\Phi}_{\mathrm{R}}\mathbf{y}_0$; with the Floquet ansatz $\mathbf{y}_p = \mu\,\mathbf{y}_0$ the multipliers follow from the generalized eigenvalue problem $\mu\,\mathbf{\Phi}_{\mathrm{L}}\mathbf{v} = \mathbf{\Phi}_{\mathrm{R}}\mathbf{v}$ at $\mathcal{O}(p)$ cost.

The present paper starts from a simple observation: *the multiplication-free form only needs a per-step residual.* Semi-discretization is, structurally, a fixed-step time integrator whose only approximation is the per-step treatment of the coefficients and the delayed term. Consequently, the piecewise-constant step can be replaced by *any* one-step scheme — an explicit Runge–Kutta method or an implicit collocation-type method — by placing the Butcher-tableau coefficients into the residual block rows and augmenting the per-step state with the internal stage values. This raises the achievable convergence order arbitrarily, at a quantifiable structural price: the sparse matrices become denser as the stage number grows. We call the resulting family the *solution-operator semi-discretization* (SOSD): the banded pair $(\mathbf{\Phi}_{\mathrm{L}}, \mathbf{\Phi}_{\mathrm{R}})$ is a sparse representation of the one-period solution operator of (1), and the per-step integrator merely selects the order of that representation.

The contributions, precisely delimited, are the following:

1. the systematic embedding of arbitrary Runge–Kutta and collocation steps (via their Butcher tableaux and continuous extensions) into the multiplication-free banded stability framework for time-periodic DDEs, including time-periodic delays (Section);
2. the explicit-versus-implicit matrix-structure and CPU trade-off analysis, and the demonstration that the single-step limit reproduces the known global collocation method (Section);
3. the identification and characterization of the accuracy–sparsity *sweet spot*: for a given accuracy target the CPU optimum lies at moderate per-step order and many steps, at a location that depends on the problem dimension and resolution demand (Section);
4. a fair-comparison methodology that separates order refinement from h-refinement and reports work–precision curves with variance on one fixed machine configuration (Section); and
5. an open-source Julia implementation, SOSD.jl, with all scripts needed to reproduce every figure.

Neither collocation itself (Breda et al. 2005) nor the multiplication-free assembly (Bachrathy 2026) is claimed as new; the value of the paper is the unifying embedding, the trade-off quantification, and the practical guidance it yields.

Background: MFSD and semi-discretization as fixed-step integration

Discretize one principal period into p uniform steps, $\Delta t = T/p$, $t_n = n\,\Delta t$, and let $r = \lceil \tau_{\max}/\Delta t \rceil$ count the steps covering the largest delay. Classical SD constructs a one-step mapping $\mathbf{x}_{n+1} = \mathbf{P}_n\mathbf{x}_n + \mathbf{R}_n\mathbf{x}_{n-r}$ from the exact solution of the frozen-coefficient system over $[t_n, t_{n+1}]$ (Insperger and Stépán 2011). The monodromy matrix is the product of p such mappings acting on the discretized history $\mathbf{y}_n = (\mathbf{x}_n^\top, \ldots, \mathbf{x}_{n-r}^\top)^\top$; forming this product scales polynomially in the resolution ($\approx \mathcal{O}(p^{2.8})$ measured in Bachrathy 2026), and its avoidance is the point of MFSD: each step contributes the residual row $\mathbf{0} = \mathbf{I}\mathbf{x}_{n+1} - \mathbf{P}_n\mathbf{x}_n - \mathbf{R}_n\mathbf{x}_{n-r}$ to a sparse banded block system over the extended vector spanning $[t - \tau_{\max},\, t + T]$, split into left/right blocks so that the spectrum follows from $\mathbf{\Phi}_{\mathrm{L}}\mathbf{y}_p = \mu\,\mathbf{\Phi}_{\mathrm{R}}\mathbf{y}_0$ via sparse Krylov iteration (Haegeman 2021) at $\mathcal{O}(p)$ total cost (Bachrathy 2026).

Two properties of this construction matter for what follows. First, the one-step map is a *fixed-step integrator*: the only approximation is that $\mathbf{A}(t)$, $\mathbf{B}(t)$ and the delayed argument are frozen (or low-order interpolated) across Δt. It is exactly this per-step treatment that caps the convergence order at two: as shown by Insperger et al. (2008), a first-order approximation of the delayed term is already optimal for the piecewise-constant coefficient scheme, and no refinement of the classical SD step exceeds order two. Second, the assembly touches the integrator only through the per-step residual; nothing in the banded structure, the left/right splitting for $p - 1 \gtrless r$, or the Krylov solver depends on *which* integrator produced the residual. The integrator is a free design choice — the theme of this paper.

Higher-order multiplication-free stepping

Stage-augmented residual blocks from a Butcher tableau

Consider an s-stage Runge–Kutta method with tableau $(\mathbf{a}, \mathbf{b}, \mathbf{c})$, a_{ij}, b_j, c_j, applied to (1) on $[t_n, t_n + \Delta t]$. Writing the stage *values* $\mathbf{Y}_i \approx \mathbf{x}(t_n + c_i\Delta t)$ (rather than stage slopes), the step reads

$$\mathbf{Y}_i = \mathbf{x}_n + \Delta t \sum_j a_{ij}\big[\mathbf{A}(t_{nj})\,\mathbf{Y}_j + \mathbf{d}_j\big], \quad i = 1, \ldots, s, \tag{2}$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t \sum_j b_j\big[\mathbf{A}(t_{nj})\,\mathbf{Y}_j + \mathbf{d}_j\big], \tag{3}$$

with $t_{nj} = t_n + c_j\Delta t$ and the delayed-plus-forcing terms

$$\mathbf{d}_j = \sum_{k=1}^{g} \mathbf{B}_k(t_{nj})\,\tilde{\mathbf{x}}\big(t_{nj} - \tau_k(t_{nj})\big) + \mathbf{c}(t_{nj}), \tag{4}$$

where $\tilde{\mathbf{x}}$ denotes the interpolated delayed state (Section). The per-step unknown block is the stage-augmented vector

$$\mathbf{w}_n = \big(\mathbf{Y}_1^\top, \ldots, \mathbf{Y}_s^\top, \mathbf{x}_{n+1}^\top\big)^\top \in \mathbb{R}^{(s+1)d}, \tag{5}$$

and the global unknown $\mathbf{q} = (\mathbf{w}_1^\top, \ldots, \mathbf{w}_p^\top)^\top$ collects all steps of one period. Equations (2)–(3) are linear in the current and delayed states, so each step contributes one block row to a sparse system $\mathbf{\Phi}_\mathrm{L}\,\mathbf{q} = \mathbf{\Phi}_\mathrm{R}\,\mathbf{q}_\mathrm{hist}$, exactly as in MFSD but with $(s+1)d$-sized blocks.

The structure of these blocks is a Kronecker (dyadic) composition of three independent ingredients: the Butcher tableau, the system matrices, and the interpolation weights. Collecting the tableau into the compound $(s{+}1) \times s$ matrix

$$\hat{\mathbf{A}} \;=\; \begin{bmatrix}\mathbf{a}\\ \mathbf{b}^\top\end{bmatrix}, \qquad \mathbf{K} = \hat{\mathbf{A}} \otimes \mathbf{I}_d, \tag{6}$$

and writing $\boldsymbol{\mathcal{A}}_n = \operatorname{diag}\big(\mathbf{A}(t_{n1}), \dots, \mathbf{A}(t_{ns})\big)$ for the stage-wise coefficient block, the step (2)–(3) reads, for the stacked unknown block $\mathbf{w}_n$,

$$\big[\mathbf{I} - \Delta t\,\mathbf{K}\boldsymbol{\mathcal{A}}_n\mathbf{S}\big]\mathbf{w}_n = \big(\mathbf{1}_{s+1} \otimes \mathbf{I}_d\big)\,\mathbf{x}_n + \Delta t\,\mathbf{K}\,\mathbf{d}_n, \tag{7}$$

where $\mathbf{S} = [\,\mathbf{I}_{sd}\;\;\mathbf{0}\,]$ selects the stage part of $\mathbf{w}_n$ and $\mathbf{d}_n$ stacks the terms (4). Each delayed sample in $\mathbf{d}_n$ is itself a dyad: stage j of delay k contributes

$$\Delta t\,\big(\hat{\mathbf{A}}\,\mathbf{e}_j \otimes \mathbf{B}_k(t_{nj})\big)\,\big(\boldsymbol{\ell}(\theta_{njk})^\top \otimes \mathbf{I}_d\big), \tag{8}$$

i.e. the outer product of a Butcher-tableau column, the delay coefficient matrix, and the continuous-extension weight row $\boldsymbol{\ell}(\theta)^\top$ evaluated at the fractional position θ_{njk} of the delayed instant. The entire assembly therefore separates into precomputable factors (tableau, weights) and pointwise coefficient evaluations $\mathbf{A}(t^*)$, $\mathbf{B}_k(t^*)$ — no matrix products between steps ever occur. Figure 1 shows the resulting sparsity patterns: $\mathbf{\Phi}_\mathrm{L}$ has identity diagonal blocks and is block-lower triangular for *both* explicit and implicit tableaux (after the per-step elimination below), so linear solves reduce to forward substitution with zero fill-in; all delay couplings that reach into the pre-period history populate $\mathbf{\Phi}_\mathrm{R}$. The per-block fill of the delay band grows with the stage number — the mechanism that will drive the accuracy–sparsity sweet spot.

In the implementation, the stage equations (2) are *pre-eliminated* per step: the stage-coupling matrix $\mathbf{M}_n = \mathbf{I} - \Delta t\,(\mathbf{a} \otimes \mathbf{I}_d)\operatorname{diag}\big(\mathbf{A}(t_{n1}), \dots, \mathbf{A}(t_{ns})\big)$ of size $sd \times sd$ is factorized once (LU; a dense inverse for small sd), and the responses of $(\mathbf{x}_{n+1}, \mathbf{Y}_1, \dots, \mathbf{Y}_s)$ to (i) $\mathbf{x}_n$, (ii) each delayed-state sample, and (iii) the additive forcing are precomputed as $(s+1)d \times d$ transition blocks. This keeps $\mathbf{\Phi}_\mathrm{L}$ *unit block-lower triangular* — its LU factorization is trivial and fill-free — while the per-step elimination costs $\mathcal{O}\big(p\,(sd)^3\big)$, linear in p. For explicit tableaux (strictly lower-triangular $\mathbf{a}$) the elimination degenerates to forward substitution; for implicit collocation tableaux (Gauss–Legendre of order $2s$, Radau IIA of order $2s-1$, Lobatto IIIA of order $2s-2$ (Hairer and Wanner 1996; Butcher 2016)) it is a genuine dense solve, which is precisely where the superconvergent order is purchased.

Delay interpolation by continuous extension

The delayed argument $t_{nj} - \tau_k(t_{nj})$ generally falls between grid nodes, inside some earlier step m. Interpolating the delayed state with an order lower than that of the integrator silently caps the observed convergence order — the higher-order analogue of the remark of Insperger et al. (2008) for the classical scheme, and a classical result of DDE numerics (Bellen and Zennaro 2003). The consistent choice is the *continuous extension* of the step itself: for collocation methods, the collocation polynomial through $\big(\mathbf{x}_m, \mathbf{Y}_{1,m}, \dots, \mathbf{Y}_{s,m}, \mathbf{x}_{m+1}\big)$ evaluated at the fractional position $\theta \in [0,1]$ of the delayed instant inside step m; for explicit methods, the scheme's dense-output polynomial (for RK4 the classical cubic Hermite-type formula). The interpolation weights depend only on θ and the tableau, so they join the precomputed transition blocks without changing the banded structure; each delayed sample couples block row n to the two neighbouring history blocks m and $m+1$. Since the delayed state lies at least one full delay in the past, these couplings stay strictly below the diagonal and $\mathbf{\Phi}_\mathrm{L}$ remains triangular; couplings reaching into the pre-period history enter $\mathbf{\Phi}_\mathrm{R}$ instead. Time-periodic delays $\tau_k(t)$ are handled naturally, because θ and the target block m are evaluated per stage from $\tau_k(t_{nj})$.

The single-step limit is global collocation

Setting $p = 1$ with an s-stage Gauss tableau makes the single step span the entire period: the stage values *are* then the collocation samples of a global polynomial, and the construction reproduces the known collocation/pseudospectral method for DDEs (Breda et al. 2005, 2015) in its strong form. This extreme is explicitly *not* new; we include it (i) to make the equivalence concrete — in Section the single-step spectrum matches the converged reference to near machine precision — and (ii) to expose its cost: the single-step matrix is effectively dense, and the number of nonzeros of the assembled system grows as $\mathcal{O}(d^2s^2)$ with the stage count (and its factorization cost as $\mathcal{O}(d^3s^3)$), so pushing all accuracy into one step abandons exactly the sparsity that makes the multiplication-free form fast. The continuum between "p large, s small" and "$p = 1$, s large" is the design space this paper maps.

Fair-comparison methodology

Comparisons between DDE stability methods frequently pit an order-refinement method (collocation at increasing polynomial order) against an h-refinement method (semi-discretization at decreasing step size) and report accuracy at a single, often low, resolution on linear axes. Such comparisons cannot identify asymptotic rates, hide constant factors, and — like comparing p-type against h-type FEM refinement — do not license a universal winner. The predecessor paper (Bachrathy 2026) already criticized sub-100-resolution, non-log-log comparisons; we extend that critique into a positive prescription:

1. **Work–precision axes.** The primary comparison is CPU time versus eigenvalue error, both logarithmic, over wide ranges; the secondary one is CPU time versus resolution p with fitted scaling exponents.
2. **Repetition-robust timing.** Every timed point is the *minimum* of repeated warm-started runs (the standard BenchmarkTools practice: the minimum is immune to garbage-collection and scheduler interference, whose one-sided spikes contaminate means); repetition counts and scatter are recorded alongside the results.
3. **Both refinement families, both complexity laws.** Every method is shown under h-refinement (fixed

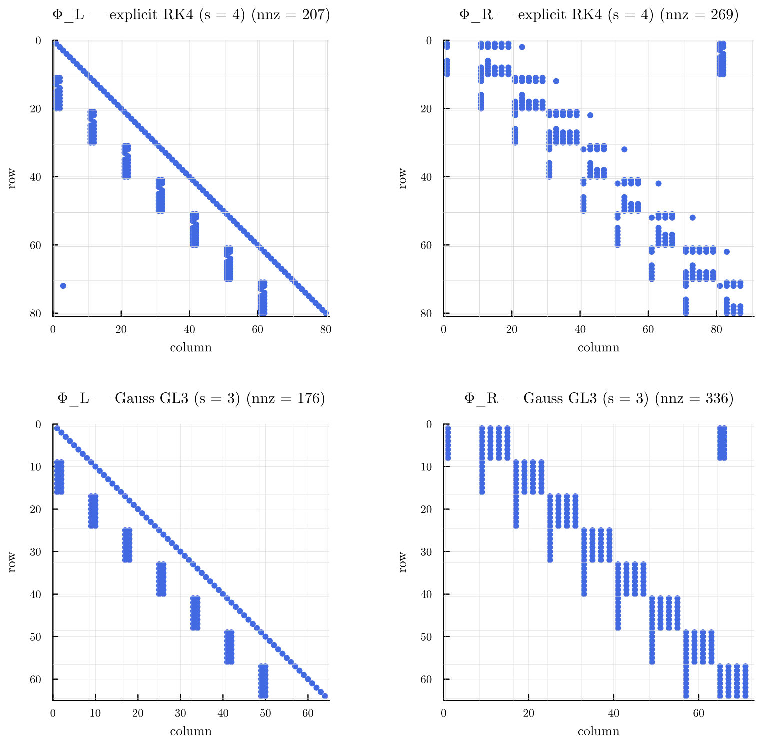

Figure 1. Sparsity patterns of the solution-operator pair $(\mathbf{\Phi}_\mathrm{L}, \mathbf{\Phi}_\mathrm{R})$ on the delayed Mathieu equation ($p = r = 8$): explicit RK4 (top) versus Gauss GL3 (bottom). $\mathbf{\Phi}_\mathrm{L}$ is unit block-lower-triangular in both cases (forward-substitution solvable, no fill-in); the delay band sits in $\mathbf{\Phi}_\mathrm{R}$, with denser per-block coupling for the collocation-type stages.

order, growing p), where the cost of the banded assembly is *linear*, $t \propto p^1$, *and* under order refinement (fixed p, growing stage number s), where the cost is *polynomial* in the order — per-step elimination $\mathcal{O}\big((sd)^3\big)$ plus stage-quadratic fill. A comparison that shows only one direction hides one of the two laws. The complete picture is the (s,p)-map of Section : error and CPU time rendered over the whole plane, with cells beyond a CPU budget left uncomputed (NaN).

4. **Fixed environment.** One machine, one BLAS thread configuration, pinned package versions, deterministic start vectors; all reported alongside the results.
5. **Verified references.** Error is measured against a reference that two distinct resolutions of a higher-order method agree on, and the reference method is cross-validated against an independent implementation (SemiDiscretizationMethod.jl).

Numerical results

Test systems

Four systems span the design space (parameters in Appendix):

1. **Delayed Mathieu equation** ($d = 2$, $\tau = T = 2\pi$): the canonical smooth validation case (Insperger and Stépán 2011).
2. **Seasonal scalar model** ($d = 1$, $\tau = 2$, $T = 2\pi$): a smooth, cheap system where high per-step order excels.
3. **Turning with spindle-speed variation** ($d = 2$, time-periodic delay, $T/\tau_\mathrm{max} \approx 9.1$): exercises the $p \gg r$ partitioning and time-varying delay handling.
4. **FEM beam with delayed boundary feedback** ($d = 28$, $\tau/T = 1/2$), optionally with act-and-wait switching of the feedback (Insperger 2006) (non-smooth stress test): the high-dimensional, high-resolution case where banded moderate order should win.

Order verification

Figure 2 and Table 1 confirm on the delayed Mathieu equation that each embedded *collocation* integrator attains its theoretical convergence order in the dominant multiplier on this *mesh-commensurate* test ($\tau = T = p\,\Delta t$): the Gauss family exhibits the superconvergent order $2s$ (fitted slopes 2.00, 4.00, 5.99 and 9.96 for $s = 1, 2, 3, 5$; GL8 reaches the 10^{-16} relative floor already at $p = 16$), Radau IIA attains $2s - 1$ and Lobatto IIIA $2s - 2$. Commensurability matters: the continuous extension of an s-stage collocation step is a degree-s polynomial with uniform accuracy $\mathcal{O}(\Delta t^{s+1})$, so for delays that fall *inside* steps classical DDE interpolation theory (Bellen and Zennaro 2003) caps the attainable order at $\min(2s,\, s{+}1)$ unless error cancellation occurs. The measurements bracket exactly this range: on the seasonal model ($\tau = 2$, never mesh-aligned) GL2 and GL3 reduce to fitted orders 2.9 and $4.0 \approx s{+}1$, while on the SSV turning system (time-varying delay) GL2 retains the full 4.0 and GL3 attains 5.2 — between $s{+}1$ and $2s$. The explicit family is capped harder: the observed order is limited by its continuous extension, so RK4 with its classical third-order dense output

Table 1. Order verification on the delayed Mathieu equation: nominal versus fitted convergence order of the dominant multiplier (log–log slope over the pre-floor error range).

Method	Nom.	Fitted	Comment
SD (order 2)	2	3.1	aligned-delay cancellation
RK2 (Heun)	2	2.0	
RK4	4	3.3	capped: 3rd-order dense output
RK5	5	2.1	capped: stage interpolant
GL1 (midpoint)	2	2.0	
GL2	4	4.0	superconvergence $2s$
GL3	6	6.0	
GL5	10	10.0	
GL8	16	—	at 10^{-16} floor, $p \geq 16$
Radau IIA, s=3	5	4.9	
Lob. IIIA, s=3	4	4.0	
Lob. IIIA, s=4	6	6.0	

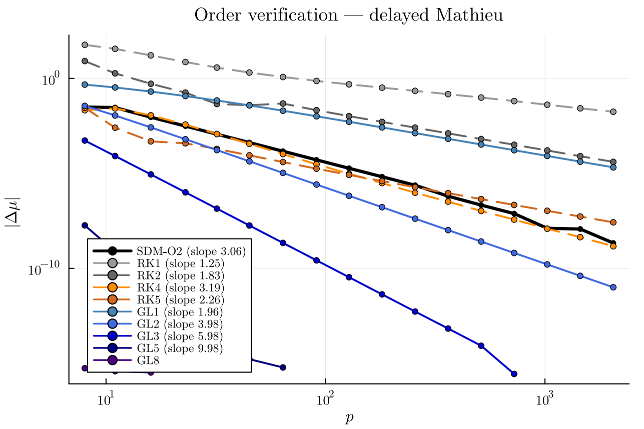

Figure 2. Order verification on the delayed Mathieu equation: eigenvalue error versus resolution p (log–log), one curve per method, dashed lines showing fitted slopes.

converges with slope ≈ 3.3 instead of 4, and RK5 with a generic stage-value interpolant collapses to slope ≈ 2.1 — the delayed lookup, not the step, is the binding constraint, exactly as anticipated in Section . Even with the $s{+}1$ cap, collocation stages carry a consistent high-order extension at no extra cost, and their nodal order can exceed it up to $2s$; sustaining comparable orders explicitly would require bespoke dense output of matching order. This asymmetry is a structural argument for implicit collocation steps inside the multiplication-free framework that the usual "explicit steps are cheaper" intuition misses.

Work–precision and time complexity

Figure 3 shows the three canonical panels for the delayed Mathieu equation: error versus p, CPU time versus p, and the work–precision plane. Every embedded integrator preserves the near-linear time complexity of the multiplication-free assembly — the fitted scaling exponents lie between 0.9 and 1.2 (median 1.05) for all methods from RK2 through GL8, rising to at most 1.6 on the high-dimensional beam — so raising the per-step order shifts the CPU-time line by an essentially constant factor (the block fill) without changing its slope. On the work–precision plane this is decisive: GL5 and GL8 reach relative errors of 10^{-14} within milliseconds, whereas the order-2 baseline spends an order of magnitude more time to reach only 10^{-6}. The same qualitative picture holds for the scalar seasonal model and for the spindle-speed-variation turning system (time-periodic delay, $p \gg r$ partitioning); the beam work–precision numbers appear in Section and Appendix .

For baseline parity with the predecessor (Bachrathy 2026), Figure 4 reproduces, in the present environment, the time-complexity contrast between the traditional monodromy construction (SD-classic, fitted exponent 2.2 over the range affordable under a 2 s-per-solve budget, $p \leq 10^3$; the predecessor measured $\approx$2.8 over a wider window) and the banded LR mapping (fitted exponent 1.05, continued here to $p = 10^6$ in 13 s); both agree in the multiplier to 10^{-15}, and at the last common resolution $p = 10^3$ the multiplication-free route is already 1200$\times$ faster.

The accuracy–sparsity sweet spot

For a fixed relative accuracy target ε, sweeping the per-step order s while choosing the smallest resolution p that reaches ε produces a U-shaped CPU-time curve: low orders need excessive p, high orders pay polynomially growing per-step cost — stage elimination $\mathcal{O}\big((sd)^3\big)$ plus fill, with measured fixed-p exponents $t \propto s^k$, $k = 1.6$ (Mathieu, $d = 2$), 1.8 (beam, $d = 28$) and 2.0 (turning at $p = 3428$) for rank-one delay coupling, and $k \to 3$ asymptotically as the stage elimination dominates. A first-order cost model, cost $\propto p\, s^k$ with $\varepsilon \propto (\Delta t/T_{\mathrm{c}})^{2s}$ for a characteristic time T_{c} (taking the best-case nodal order $2s$; reduced orders for non-commensurate delays shift the optimum mildly upward, and the factorial decay of the Gauss error constants is neglected), predicts the optimum

$$s^* \approx \tfrac{1}{2k} \ln(1/\varepsilon), \tag{9}$$

i.e. even a 10^{-12} accuracy target asks for only $s^* \approx$ 5–10 Gauss stages — moderate order, many steps. This asymptotic model applies when the fill actually binds. For the small, smooth Mathieu system it does not: the measured optima sit at $s^* =$ 5–8 with as few as $p^* =$ 2–6 steps (e.g. $\varepsilon = 10^{-12}$ in 4 ms with $s = 8$, $p = 6$, $\mathrm{nnz}(\boldsymbol{\Phi}_{\mathrm{L}}) = 288$) — for low-demand problems the whole matrix is tiny and the near-spectral corner is optimal. The fill term takes over exactly in the contrasting cases: the measured per-step fill grows from 12 nonzeros at $s = 1$ to 112 at $s = 20$ on the Mathieu system but from $1.7 \cdot 10^3$ ($s = 1$) to $6.4 \cdot 10^3$ ($s = 6$) per step on the $d = 28$ beam, whose optima correspondingly retreat to $s^* =$ 4–5 (relative accuracy 10^{-8} in 11 ms at $s = 5$, $p = 6$; targets beyond 10^{-8} are excluded as they approach the reference accuracy).

Figure 5 renders the full picture: relative error and CPU time as colour maps over the (s, p) plane, with the two complexity laws visible as the gradient directions — along p at fixed s the cost grows linearly, along s at fixed p polynomially — and the per-target optima marked. Cells whose cost exceeds the CPU budget are not computed (blank). For a given error contour, the cheapest cell is never the single-step/ultra-high-order corner once the resolution demand or the system dimension is nontrivial: the polynomial s-law meets the linear p-law at moderate order.

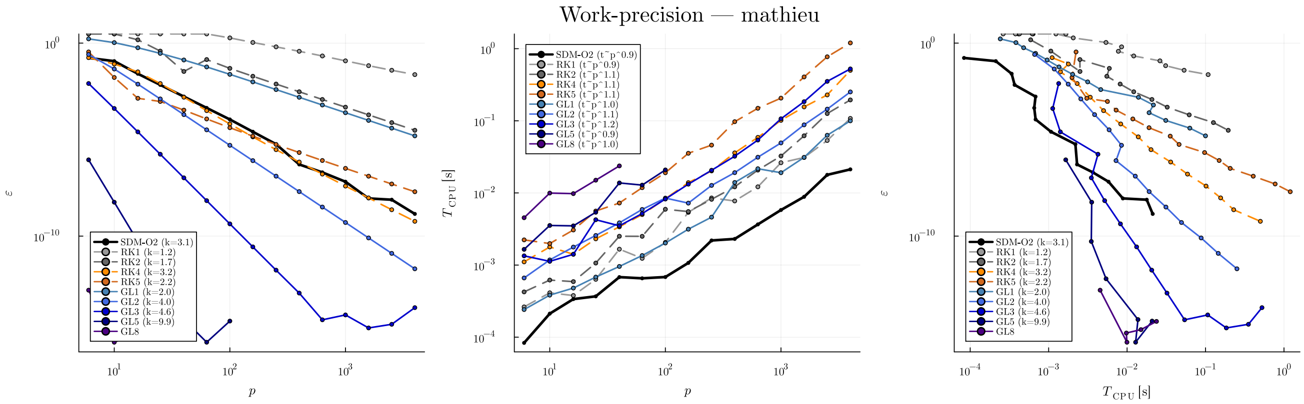

Figure 3. Work-precision study on the delayed Mathieu equation. Left: eigenvalue error vs. resolution p (clipped at $3 \cdot 10^0$ — pre-resolution values carry no information); middle: CPU time vs. p, all fitted exponents ≈ 1.0; right: the work–precision plane (down and left is better). CPU times are minima of repeated warm-started benchmark measurements; their relative scatter is below a few per cent and is therefore not drawn.

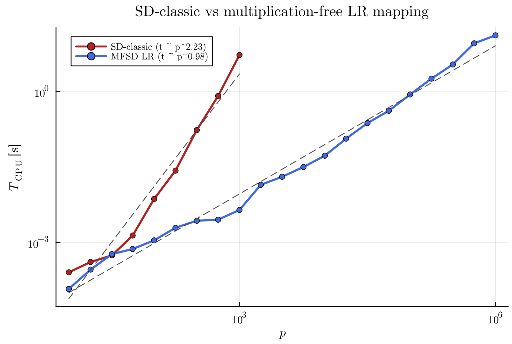

Figure 4. Baseline parity with the predecessor: CPU time of the traditional monodromy build (SD-classic, capped at 2 s per solve) versus the banded multiplication-free mapping (continued to $p = 10^6$) on the delayed Mathieu equation; minimum of repeated warm-started measurements, dashed lines: fitted scaling.

Contrasting resolution demands: where the banded route wins

The location and the practical consequences of the sweet spot depend on how many points per period the problem physically requires. Two showcases bracket the range:

- **Low demand (tens of points).** For the delayed Mathieu equation and the seasonal scalar model, resolutions of $p \approx 10$–30 already reach high accuracy with moderate Gauss orders; matrices are so small that sparsity is irrelevant, and even the single-step spectral corner is viable.
- **High demand (hundreds of points and more).** The spindle-speed-variation turning system oscillates with $T/T_{\rm osc} \approx 36$ (natural frequency high relative to the period), so *any* method needs $N = p(s{+}1) \gtrsim 500$ samples per period to resolve the dynamics at all. Here the banded moderate-order route wins decisively at engineering accuracies: the measured optima are $(s^*, p^*) = (7, 47)$ for $\varepsilon = 10^{-4}$ (21 ms), $(5, 159)$ for 10^{-6} (37 ms) and $(7, 100)$ for 10^{-8} (49 ms) — tens to hundreds of steps at moderate order, with the sample count pinned near the bound ($N \approx 380$–950). Only at the extreme target $\varepsilon = 10^{-12}$ does the optimum shift towards high order, $(20, 22)$ at 91 ms — and even there $N = 462$ respects the bound and the step count remains far from the single-step corner, which is not computable within the CPU budget at all.

The resolution demand can be estimated a priori by a sampling (Shannon-type) argument. The discretization carries $N = p\,(s{+}1)$ solution samples per period; resolving the fastest natural oscillation of the system, $\omega_{\max}$, requires at least the Nyquist floor of two samples per oscillation period and, for engineering accuracy, empirically $c_{\rm s} \approx 5$–10 samples:

$$N \;=\; p\,(s{+}1) \;\gtrsim\; c_{\rm s}\,\frac{\omega_{\max} T}{2\pi}. \qquad (10)$$

For the turning system ($\omega_{\max} T/2\pi \approx 36$) this predicts $N \gtrsim$ 200–400, matching the observed onset of convergence; below it the computed multipliers are meaningless (for both SOSD–GL2 and the order-2 baseline at $p = 50$, i.e. $N = 150$, they are off by orders of magnitude). The bound (10) sharpens the sweet-spot analysis: when N is pinned by the oscillation content rather than by the accuracy target, the cost of the banded route scales as $p\,s^k \approx N\,s^{k-1}$ — *still increasing in the order at fixed sample count* for every measured exponent k — so the smallest stage number that meets the accuracy target is CPU-optimal, and the single-step spectral corner (all N samples in one dense block) is maximally penalized. Conversely, when the problem is easy (N shrinks exponentially with growing s), higher order pays until the fill takes over. The same contrast governs high-dimensional systems: for the FEM beam ($d = 28$), memory — the per-step fill — caps the affordable stage number before CPU time does.

An important caveat on superconvergence follows. Figure 6 shows the order-refinement direction explicitly: the error *and* the CPU time versus the stage number s (up to $s = 100$) at fixed step counts $p = 1, 10, 100$, for the easy and the hard system. Superconvergence does *not* mean that a handful of stages always suffices: below the resolution bound (10) the discrete spectrum is simply wrong (relative errors of order one), and the celebrated steep decay only begins once $N = p(s{+}1)$ crosses the bound — the cliff sits at $s^{\rm cliff} \approx c_{\rm s}\,\omega_{\max} T/(2\pi p) - 1$, i.e. its position is set by the natural-oscillation content of the system divided by

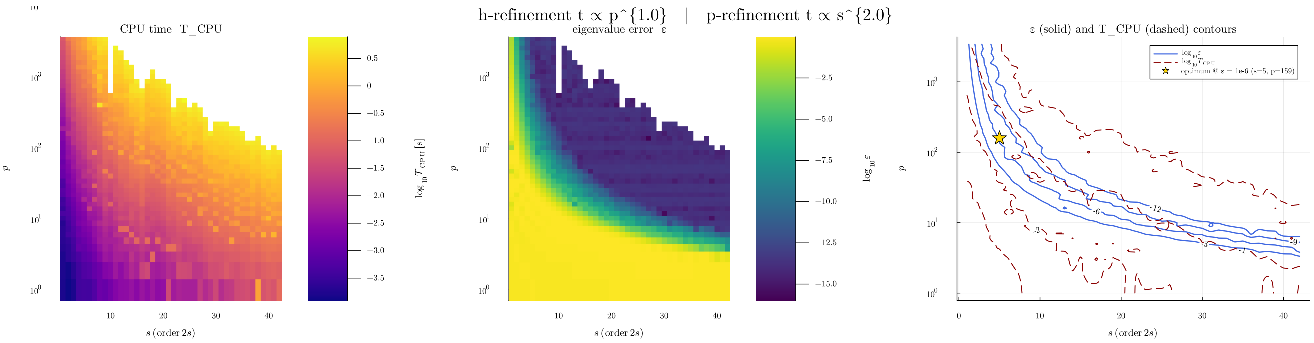

Figure 5. The (s,p)-map for the high-resolution-demand turning system ($p = 1\dots4\cdot10^3$, $s = 1\dots30$; beyond $s\approx30$ the product-form evaluation of the continuous-extension weights loses floating-point accuracy — a barycentric implementation would extend the range, which is irrelevant in practice since all optima lie at $s\le8$). Left: colour encodes CPU time; middle: colour encodes the eigenvalue error; right: both contour families overlaid — following, e.g., the $\varepsilon=10^{-6}$ error contour (solid), the smallest CPU-time contour (dashed) it touches identifies the optimal setting, marked by the star. h-refinement (vertical) is linear in cost, order refinement (horizontal) polynomial; blank cells exceed the 5 s CPU budget. The corresponding maps for the low-demand Mathieu system are in Appendix .

the step count, not by the method order. For the Mathieu system the cliff sits at $N\approx10$–30; for the turning system at $N\approx200$–400, so a single-step scheme needs s in the hundreds before it produces anything meaningful. Raising the order buys nothing until the sampling limit is met; beyond the limit it buys everything. The bottom row displays the measured CPU-versus-s law at fixed p with fitted exponents, answering the complexity question directly: the cost is polynomial in s throughout, with the exponent growing towards the cubic stage-elimination limit at large s. The order-verification and work–precision figures (error versus p at fixed s, Figures 2 and 3, and Appendix for the hard case) show the complementary h-refinement direction of the same surface.

Explicit versus implicit steps

The conventional trade-off — explicit steps are cheap but order-capped, implicit stages are expensive but superconvergent — is reshaped by two findings. First, after per-step pre-elimination *both* families yield a unit block-lower-triangular $\mathbf{\Phi}_{\mathrm{L}}$, so the downstream solve cost differs only through the block fill; the explicit advantage reduces to the cheaper (forward-substitution) per-step elimination, which is a linear-in-p term either way. Second, the delay-interpolation order requirement (Table 1) penalizes exactly the explicit family: sustaining order $q>3$ explicitly requires bespoke dense output of order q, i.e. additional stages and fill, whereas the collocation families carry their matching-order continuous extension for free. In the measured work–precision curves the explicit methods are therefore competitive only in the low-accuracy regime ($\varepsilon\gtrsim10^{-4}$), and Gauss stages dominate everywhere beyond it. On the FEM beam the contrast is stark: GL3 reaches $\varepsilon=10^{-4}$ in 6 ms and 10^{-6} in 11 ms, whereas the order-2 baseline needs 140 ms and 795 ms (23×–72×), explicit RK5 needs 3.9 s for 10^{-6} (≈360× slower than GL3), and $\varepsilon=10^{-8}$ is reached by the Gauss variants alone — the explicit family is additionally barred from coarse grids by its Courant–Friedrichs–Lewy (CFL)-like stability bound on this stiff system (Appendix).

Remark on scope: smooth systems

The high-order convergence results above — like those of every spectral or high-order method — presuppose (piecewise-)smooth coefficients with any discontinuities aligned to grid nodes. For genuinely non-smooth systems (act-and-wait switching, interrupted cutting) *all* methods of any nominal order fall back to low observed order when the discontinuity falls inside a step; this is verified for the act-and-wait beam in Appendix . Two practical consequences: the sweet-spot recommendation of moderate order with many steps holds *a fortiori* (the sparsest variants lose nothing where high order cannot pay off), and the linear-complexity advantage of the solution-operator assembly is unaffected. A discontinuity-aware treatment (mesh alignment, step splitting) is outside the scope of this paper.

Discussion: practical guidance

The measurements condense into a short decision rule for practitioners — an a priori recipe requiring only the fastest natural frequency $\omega_{\max}$ of the system, the period T, and the accuracy target ε:

1. **Resolution bound:** $N_{\min}\approx c_{\mathrm{s}}\,\omega_{\max}T/(2\pi)$ with $c_{\mathrm{s}}\approx5$–10 (Eq. (10));
2. **Stage number:** $s\approx\max\bigl(2,\ \lceil\ln(1/\varepsilon)/(2k)\rceil\bigr)$ with the measured cost exponent $k\approx1.6$–2.0 (Eq. (9)), capped by memory for large d;
3. **Step count:** $p=\lceil N_{\min}/(s{+}1)\rceil$, then one or two h-refinements to confirm the target — cheap, since the cost is linear in p.

In more detail:

1. **Default: Gauss stages, moderate order.** The stage-number rule of step 2 above — $s\approx\lceil\ln(1/\varepsilon)/(2k)\rceil$ with the measured $k\approx1.6$–2.0 — gives $s=7$–9 even for $\varepsilon=10^{-12}$, in agreement with the measured optima $s^*=4$–8 at engineering targets across all systems. Then pick p by one or two refinement steps until the target is met — the cost is linear in p, so overshooting p is cheap, whereas overshooting s inflates the per-step cost polynomially.

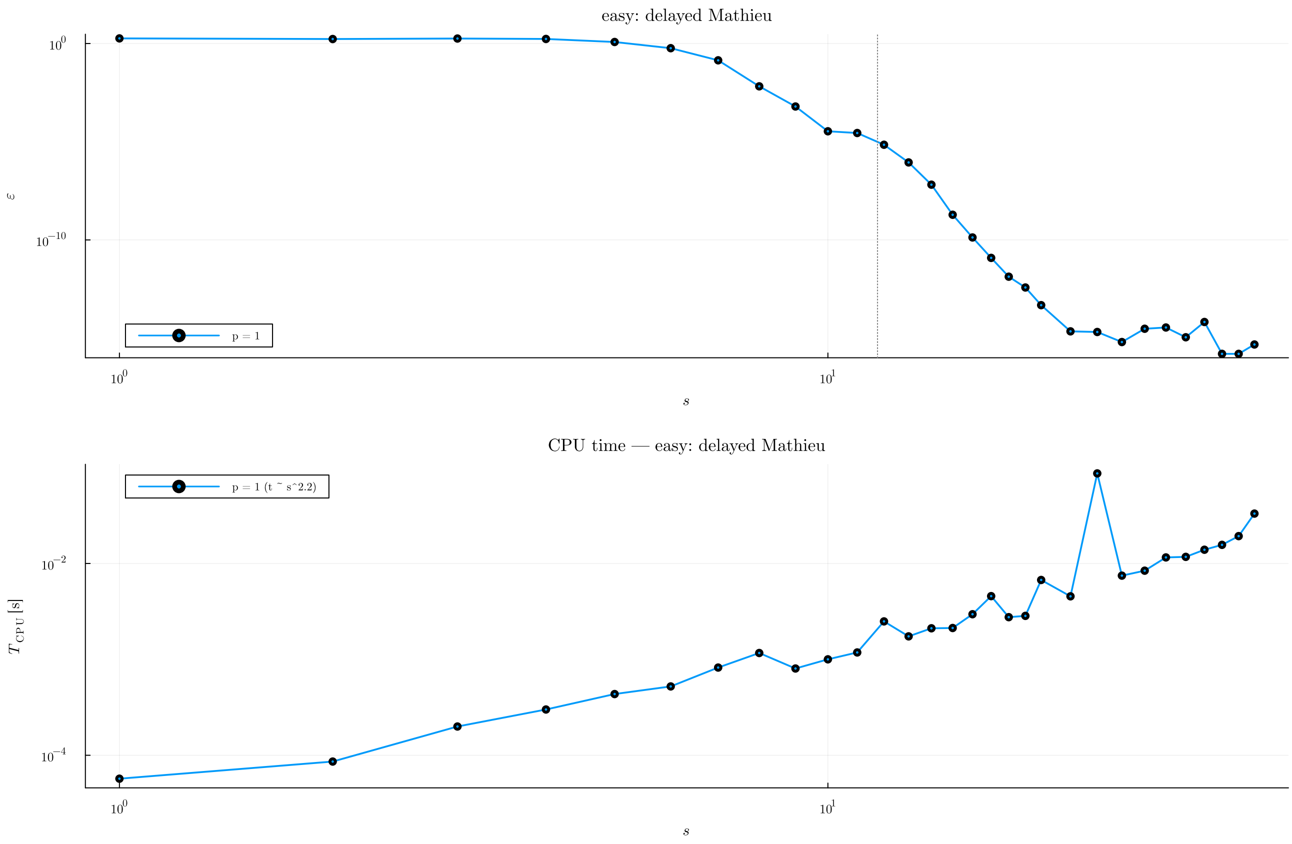

Figure 6. Order refinement at fixed step counts $p = 1, 10, 100$ for the easy (left) and hard (right) system. Top: relative error versus stage number s (clipped at $3 \cdot 10^0$); dotted vertical lines mark the sampling estimate (10). Superconvergent decay starts only beyond the bound, whose position scales with $\omega_{\max} T/(2\pi p)$. Bottom: the measured CPU time versus s with fitted large-s exponents (minimum of repeated measurements).

2. **Large systems: cap the order earlier.** The per-step fill grows with d^2 and polynomially with s; for the FEM-scale beam ($d = 28$) the measured optima sit at $s^* = 4$–5 with $p^* \leq 6$, and memory, not CPU, becomes the binding constraint for larger stage numbers.
3. **Avoid the spectral corner for engineering models.** The single-step (global collocation) mode is elegant for small, smooth, low-accuracy-run-count problems, but it abandons sparsity (nnz $\sim d^2 s^2$ in one dense block, factorization $\sim d^3 s^3$) and loses its order advantage entirely for non-smooth coefficients (act-and-wait, interrupted cutting).
4. **Explicit steps only for cheap low-accuracy sweeps on non-stiff systems.** Their observed order is capped by the available dense output (≈ 3 for classical RK4), and on stiff high-dimensional systems they are unusable below a CFL-like resolution bound, where the A-stable Gauss stages converge from the coarsest grids.
5. **Non-smooth coefficients: align or refine.** If switching instants can be placed on grid nodes, full order is retained; if not, expect order collapse and prefer moderate order with more steps.

Conclusion

Once the stability analysis of time-periodic DDEs is written in multiplication-free banded form, the per-step integrator is a free design choice: Butcher-tableau residual blocks with continuous-extension delay interpolation raise the convergence order to that of the chosen scheme, up to Gauss superconvergence $2s$, while retaining $\mathcal{O}(p)$ complexity. The two classical method families are the endpoints of one continuum — piecewise-constant semi-discretization at maximal sparsity and global collocation at maximal per-step order — and neither endpoint is optimal: the CPU-time optimum sits at a problem-dependent sweet spot of moderate order and many steps, drifting only logarithmically with the accuracy target and towards lower order for larger system dimension and non-smooth coefficients. This unifies and clarifies, rather than reinvents; the actionable outputs are the sweet-spot guidance and the open-source SOSD.jl implementation.

Declaration of conflicting interests

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Model parameters

Delayed Mathieu equation $(d = 2)$

$$\begin{aligned}\ddot{x} + a_1\dot{x} + \big(\delta + \epsilon_{\mathrm{M}}\cos(2\pi t/T)\big)x \\ = b_0\, x(t-\tau) + \sin(4\pi t/T),\end{aligned}$$

with $\delta = 3$, $\epsilon_{\mathrm{M}} = 0.2$, $b_0 = -0.15$, $a_1 = 0.1$, $\tau = T = 2\pi$. Converged dominant multiplier $\mu = 0.351417260953549$ (two-resolution GL10 reference, Appendix).

Seasonal scalar model $(d = 1)$ $\dot{y}(t) = -(a + b\cos t)\, y(t-\tau)$ with $a = b = 1$, $\tau = 2$, $T = 2\pi$; $\mu = 0.626021929290183$.

Turning with spindle-speed variation $(d = 2)$

$$\begin{aligned}\ddot{x} + \zeta\dot{x} + (1 + k_w)x = k_w\, x\big(t - \tau(t)\big) + \cos(2\pi t), \\ \tau(t) = \frac{2\pi}{\Omega}\Big(1 + A_{\mathrm{SSV}}\sin\tfrac{2\pi t}{T}\Big),\end{aligned}$$

with $k_w = 0.2$, $\zeta = 0.1$, $\Omega = 0.3$, $A_{\mathrm{SSV}} = 0.1$, $T = N_T\, 2\pi/\Omega$, $N_T = 10$ (so $T/\tau_{\max} \approx 9.1$); $\mu \approx 4.47617$.

FEM beam with delayed boundary feedback $(d = 28)$ Longitudinal rod of length $L = 100\,\mathrm{m}$, Young's modulus $E = 210\,\mathrm{GPa}$, cross-section $A = 10^{-4}\,\mathrm{m}^2$, density $\rho_{\mathrm{m}} = 7800\,\mathrm{kg/m}^3$, stiffness-proportional damping $\eta = 0.01$, discretized by $N_{\mathrm{e}} = 15$ linear elements with lumped mass, fixed at the first node ($d = 2(N_{\mathrm{e}} - 1) = 28$ states). Delayed displacement feedback from the first free node to the last node with gain $P\,E/(L/N_{\mathrm{e}})$, $P = 0.2$; wave speed $c_0 = \sqrt{E/\rho_{\mathrm{m}}}$, $T_{\mathrm{sp}} = L/c_0$, delay $\tau = 0.2\,T_{\mathrm{sp}}$, period $T = 0.4\,T_{\mathrm{sp}}$ ($\tau/T = 1/2$); constant unit force at the free end. In the act-and-wait configuration the feedback matrix is active only for $t < 0.8\,T$. Reference multipliers: $\mu = 373.008322$ (smooth) and $\mu = 319.636074$ (act-and-wait, aligned-grid reference).

Additional work–precision studies

Figures 8–10 repeat the three-panel work–precision study of Figure 3 for the remaining test systems. The qualitative picture is identical: linear time complexity for every embedded integrator, Gauss stages dominating beyond $\varepsilon \approx 10^{-4}$. On the stiff beam the explicit methods are unstable below a CFL-like resolution bound (errors up to 10^{43}), while the A-stable Gauss stages converge from the coarsest grids. Figure 7 shows the corresponding (s, p)-maps for the low-demand Mathieu system.

Non-smooth coefficients: act-and-wait beam

With act-and-wait switching, the beam's feedback matrix jumps at $t = 0.8\,T$. On grids where the switching instant is a node (p divisible by 5) the nominal order is retained (measured slopes 2.0 for GL1 and 3.7 for GL2; GL3 sits at the reference floor from the coarsest aligned grid). On off-grid meshes (powers of two) the observed order of *every* method collapses to ≈ 1 (measured 1.2, 1.1, 0.9 for GL1–GL3; Figure 11), confirming the scope remark of Section : with unaligned non-smoothness, all methods fall back to first order, so nothing is lost by choosing the sparse moderate-order variants.

Brute-force stability charts at the optimal setting

As an end-to-end demonstration of the sweet-spot guidance, high-resolution stability charts are computed by brute force — one eigenvalue solve per grid point (each solve single-threaded), using the CPU-optimal setting of each system. Figure 12 shows the results, coloured by the logarithm of the spectral radius (blue: stable, red: unstable); the stability boundary $\rho = 1$ is refined to sub-cell accuracy by the multi-dimensional bisection method (MDBM, Bachrathy and Stépán 2012) on top of the colour map. The grid evaluation is trivially parallel (every point is an independent solution-operator solve) and is executed on 16 threads. The delayed Mathieu chart ($100 \times 100 = 10000$ points, GL7, $p = 2$) completes in 3 s wall time plus 3 s for the MDBM boundary; the strongly modulated SSV turning chart ($100 \times 100 = 10000$ points, GL4, step count chosen per point by the sampling recipe — a resolution-pinned hard system) in 55 s wall time plus 22 s for the boundary. No continuation tricks are used; the per-point cost is simply the optimally chosen solution-operator solve.

Structural remarks: delay–period ratios, sub-step delays, and mass matrices

Period-to-delay ratio. The ratio $T/\tau_{\max}$ (i.e. p versus r) never affects the triangularity of $\mathbf{\Phi}_{\mathrm{L}}$; it only selects the splitting into the left/right pair (the $p - 1 \gtrless r$ partitioning of Bachrathy 2026), and for $\tau_{\max} > T$ the assembly simply wraps the overlapping part of the history through unchanged — still triangular. The turning example ($T/\tau_{\max} \approx 9.1$) exercises the $p \gg r$ case throughout this paper.

Sub-step delays. What *does* change the structure is the delay relative to the *step*: for $\tau_k(t) \geq \Delta t$ every delayed lookup references strictly earlier blocks and $\mathbf{\Phi}_{\mathrm{L}}$ is unit block-lower-triangular. For $\Delta t > \tau_k$ stage lookups reach into the *current* step, so the diagonal block becomes a non-identity but small and invertible matrix: block forward substitution still applies at linear cost, although even an explicit tableau then requires a per-step solve (the classical "overlapping" phenomenon of DDE integration). Entries *above* the diagonal would require advanced-type arguments ($\tau < 0$) and never arise for retarded systems.

Mass matrices and differential–algebraic systems. The descriptor form $\mathbf{M}\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x} + \sum_k \mathbf{B}_k\mathbf{x}(t - \tau_k) + \mathbf{c}(t)$ replaces the identity blocks of the stage system (7) by $\mathbf{M}$:

$$\Big[\mathbf{I}_s \otimes \mathbf{M} - \Delta t\,(\mathbf{a} \otimes \mathbf{I}_d)\,\boldsymbol{\mathcal{A}}_n\Big]\mathbf{Y} = (\mathbf{1}_s \otimes \mathbf{M})\,\mathbf{x}_n + \dots \tag{11}$$

For regular $\mathbf{M}$ this is a drop-in replacement (the per-step LU absorbs $\mathbf{M}$; no explicit inverse is ever formed). For *singular* $\mathbf{M}$ — a delay differential–algebraic system — the update row would require $\mathbf{M}^{-1}$, unless the tableau is *stiffly accurate* ($c_s = 1$, $b_j = a_{sj}$: Radau IIA, Lobatto IIIA), in which case $\mathbf{x}_{n+1} = \mathbf{Y}_s$ and no inversion occurs; the generalized eigenvalue formulation also absorbs the infinite spectrum of the algebraic part naturally. As a minimal validation (implemented in `examples/dae_mass_matrix.jl`), the index-1 system

$$\dot{x}_1 = -x_1 + x_2 + \tfrac{1}{2}x_1(t-1), \qquad \nu\dot{x}_2 = \sin(t)\,x_1 - x_2,$$

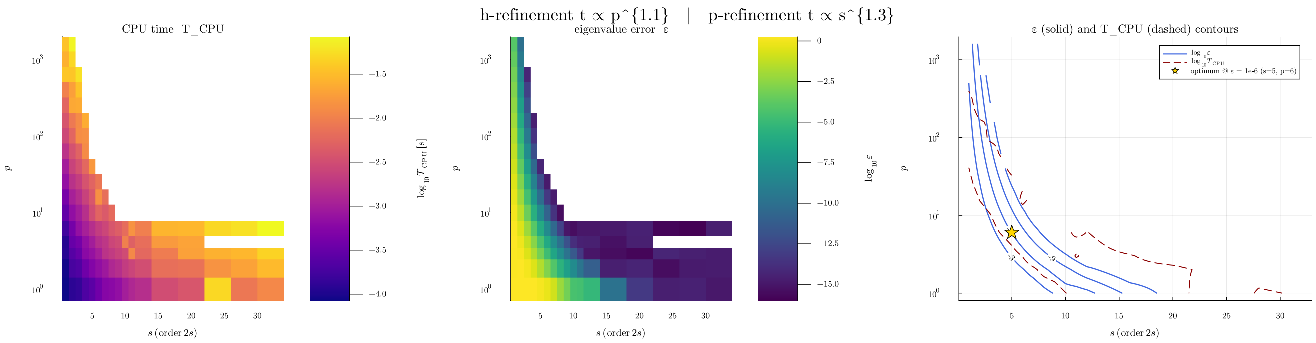

Figure 7. The (s, p)-map for the low-demand delayed Mathieu equation, with the same panel logic as Figure 5 (colour = CPU time with labelled error contours; inverse on the right): here deep accuracy is reached in the high-order/low-p region as well, and the optima sit near the spectral corner because the matrices stay tiny; the polynomial CPU growth along s is nevertheless visible at the extended stage orders.

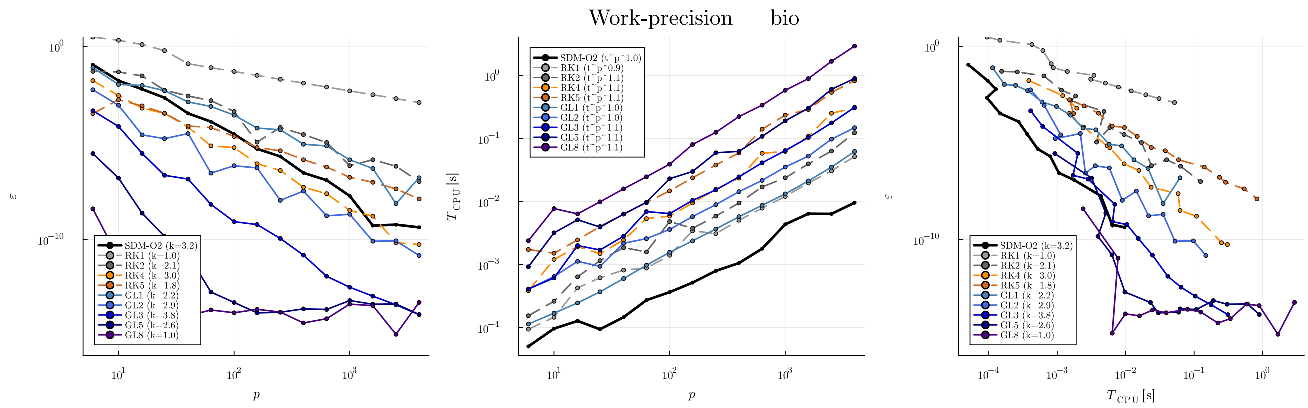

Figure 8. Work–precision study for the seasonal scalar model; same panel logic as Figure 3, parameters in Appendix .

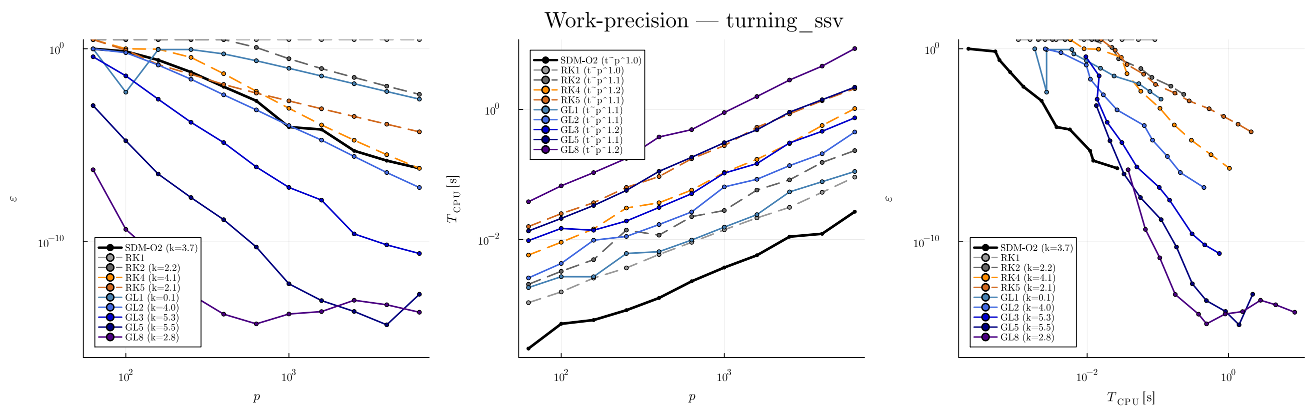

Figure 9. Work–precision study for the turning system with spindle-speed variation; same panel logic as Figure 3, parameters in Appendix .

with $\mathbf{M} = \mathrm{diag}(1, \nu)$, reduces for $\nu = 0$ to a scalar linear DDE whose multiplier serves as reference. Radau IIA ($s = 3$) applied to the *singular* problem converges to this reference at observed order $\approx 4 = s+1$ — consistent with the delay-interpolation cap of Section on this non-commensurate delay — (relative errors $3.8 \cdot 10^{-8}$, $4.0 \cdot 10^{-9}$, $3.4 \cdot 10^{-10}$, $1.6 \cdot 10^{-11}$ for $p = 25, 50, 100, 200$), and the singularly perturbed case $\nu = 10^{-6}$ converges stably to the DAE limit within the expected $\mathcal{O}(\nu)$ offset ($\approx 10^{-6}$). A full convergence theory for delay DAEs is beyond the scope of this paper.

Reproducibility

All computations were performed with Julia 1.12.4 on Windows 11 (x86-64), with BLAS restricted to a single thread, package versions pinned by the repository manifests (SOSD.jl v0.1.1 with KrylovKit, LinearMaps, RungeKutta, StaticArrays; baseline SemiDiscretizationMethod.jl v0.4.2). Timings are means over repeated warm-started runs reported as minima (scatter recorded in the results files); start vectors of all Krylov iterations are deterministic. Reference multipliers are accepted only when two resolutions of a GL(10) (small systems) or lazy GL(5) (beam) computation agree to 10^{-10} relative, and the reference method is

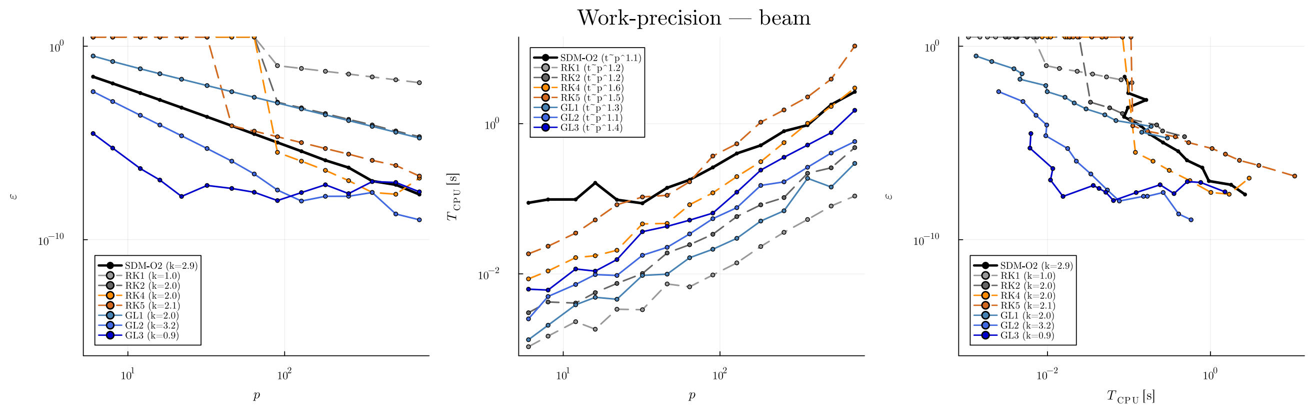

Figure 10. Work-precision — FEM beam with delayed boundary feedback. The error axis is clipped at $3 \cdot 10^0$: below the CFL-like stability bound the explicit methods produce meaningless values (up to 10^{43}), which carry no information beyond "unstable". CPU times are means of repeated benchmark measurements (scatter below a few per cent).

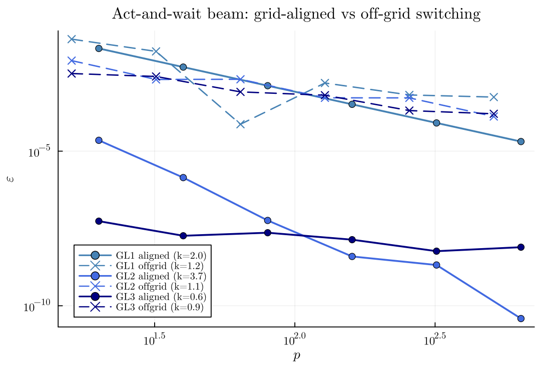

Figure 11. Act-and-wait beam: relative error versus p for GL1–GL3 on grid-aligned (solid) and off-grid (dashed) meshes.

cross-validated against SemiDiscretizationMethod.jl on every system. Every figure can be regenerated via `julia --project=benchmark benchmark/run_all.jl` followed by `julia --project=benchmark benchmark/make_figures.jl`. Repository: `https://github.com/bachrathyd/SOSD.jl` [insert commit hash at submission].

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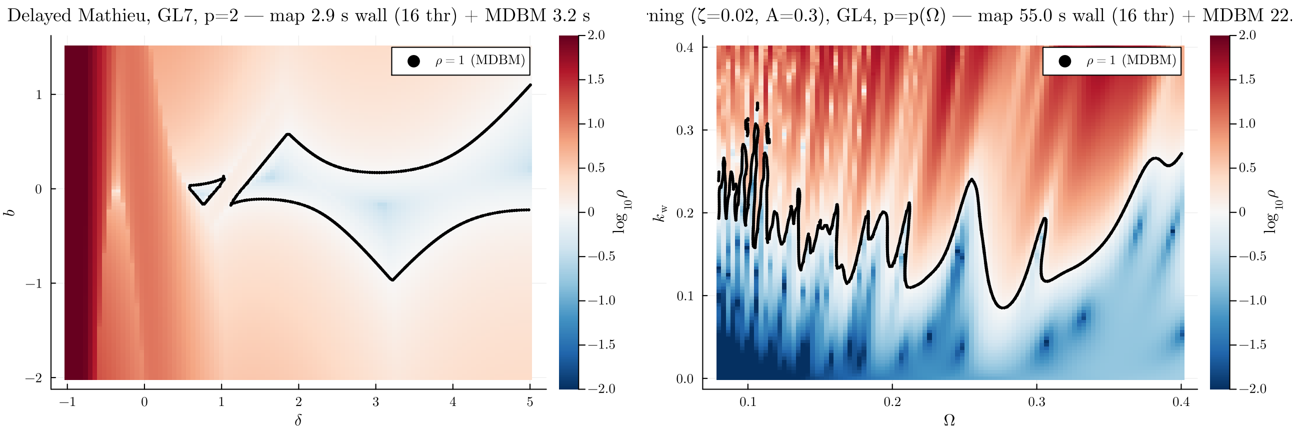

Figure 12. Brute-force stability charts at the optimal settings, coloured by $\log_{10}\rho$ (blue: stable, red: unstable); black dots: the $\rho = 1$ boundary refined by MDBM; the measured CPU times appear in each title. Left: delayed Mathieu equation in the (δ, b_0) plane ($\epsilon_{\mathrm{M}} = 1$, $a_1 = 0.1$, $\tau = T = 2\pi$), GL7 with $p = 2$. Right: SSV turning in the (Ω, k_w) plane with strong modulation ($\zeta = 0.02$, $A_{\mathrm{SSV}} = 0.3$, $N_T = 10$); here the step count follows the sampling recipe of Section per point, $p(\Omega) = \lceil 16/\Omega \rceil$ — the paper's own rule applied in a realistic production setting. Grid evaluation on 16 threads (wall times in the titles); the MDBM refinement is serial.